

Invited Article

Synthesis, growth and characterization of guanidinium hippurate monohydrate single crystals: A new third order nonlinear optical material

Jerin Susan John^a, T. Arumanayagam^b, P. Murugakoothan^b, D. Sajan^{a,*}, Nithin Joy^c, Reji Philip^c

^a Centre for Advanced Functional Materials, Post-graduate and Research Department of Physics, Bishop Moore College, Mavelikara, Alappuzha, Kerala, 690110, India

^b PG and Research Department of Physics, Pachaiyappa's College, Chennai, 600030, India

^c Light & Matter Physics Group, Raman Research Institute, Bengaluru, 560080, India

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ABSTRACT

Single crystals of a novel organic nonlinear optical (NLO) material-guanidinium hippurate monohydrate (GuHM) -are grown using slow evaporation solution growth technique. GuHM is an anion-cation system to which water molecules are attached by hydrogen bonding. The lattice parameters and crystalline purity of the crystal have been measured using single-crystal and powder X-ray diffraction (XRD) methods. Quantum chemical calculations using density functional theory (DFT) have been employed to calculate the vibrational modes and the first and second hyperpolarizabilities. Thermogravimetric (TG) and differential thermal (DT) analyses were carried out to characterize the thermal behaviour of the grown crystals. The role of intramolecular charge transfer and hydrogen bonds in making the molecule NLO-active are interpreted on the basis of the hyperpolarizability values. Along with DFT calculations, the natural bond orbital (NBO), the Atoms In Molecules (AIM), and the Hirshfeld surface imply the presence of a strong intra- and intermolecular N-H $\cdots$ O and O-H $\cdots$ O hydrogen bonding interactions between the charged species of GUHM crystal. The third-order optical nonlinearity of the crystal was studied using open-aperture Z-scan technique. From the experimental and theoretical data, it is concluded that GuHM is a potential NLO material.

1. Introduction

In the search for novel nonlinear optical (NLO) materials, organic single crystals are still an area of active research for their use in optoelectronic and photonic devices. Several carboxylate acid crystals, especially amino acid single crystals are reported to have large nonlinear optical parameters due to the presence of both donor and acceptor groups in the same structure [1–5]. Hippuric acid (also called as *N*-Benzoyl glycine) is an organic NLO carboxylic acid crystal, which crystallizes in orthorhombic structure with four molecules in a unit cell [6–9]. The growth of pure and doped hippuric acid NLO crystals have been described in literature [8,10–12] and the grown crystals are reported to have NLO properties. The attachment of a suitable cation to the acid anion forms an extended π -electron system which in turn extends the charge transfer throughout the system. Guanidine [13] is a strong base which reacts easily with organic acids to form crystals of good quality. The guanidinium cation [14] has the peculiarity that it consists of three equivalent CN bond lengths and N–C–N bond angles

which give three-fold symmetry to the structure [15]. The NH₂ groups and the electron-acceptor π -bond character of the C–N bonds make guanidinium a polarizable acentric two-dimensional cation. Studies on guanidinium substituted compounds [16–18] show that some of the compounds possess better second order NLO activity than that of the standard KDP crystal. The combination of phosphorous acid and orthoarsenic acid with guanidine form a hydrogen-bonded complex and its second harmonic activity has been reported [19]. The hydrogen bond that exists between the anion-cation system [20] significantly influences the NLO properties since it changes the electronic distribution of atoms that are involved in hydrogen bonding [21–24].

The combination of hippuric acid and guanidine forms a new hydrogen-bonded anion-cation system which is named as guanidinium hippurate monohydrate (GuHM). A thorough study including the synthesis, growth, structural, and vibrational spectral characterizations supplemented by density functional theory (DFT) calculations of the novel NLO compound, GuHM, are carried out in this work. The studies elucidate the correlation between the molecular structure and NLO

* Corresponding author.

E-mail address: drsajanbmc@gmail.com (D. Sajan).

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property by investigating the intramolecular charge transfer (ICT) interaction, hydrogen bonding, and the static first and second hyperpolarizabilities of the title compound.

2. Experimental details

2.1. Synthesis and crystal growth

GuHM crystals were synthesized using high purity guanidinium carbonate and hippuric acid in the stoichiometric ratio (1:2) according to the following reaction:

The calculated amounts of guanidinium carbonate (10.02 g) and hippuric acid (19.94 g) were dissolved separately in 20 mL of water and methanol respectively and the mixture was refluxed for 2 h. The solvent was then removed and the resultant colourless crystalline powder was purified by successive recrystallization method.

2.2. Characterization techniques

The crystal data and the structure refinement parameters for GuHM crystal were obtained from single crystal X-ray diffraction (SCXRD) study using Bruker Kappa APEX II CMOS diffractometer, while the powder XRD pattern of GuHM were collected using Rigaku SmartLab powder X-ray diffractometer.

Thermal studies by TG/DTA technique were performed by the TA instrument SDT Q600 V20.9 using nitrogen atmosphere in the range 50–900 °C at a rate of 20 °C/min. The dielectric studies were done by CV measurement using Agilent 4294A impedance analyzer in the range 40 Hz to 5 MHz. UV–Vis spectrum was recorded in the range 187–600 nm using JASCO spectrophotometer. SEM-EDS analyses of GuHM were conducted using Zeiss Ultra 55 Scanning Electron Microscope. The FTIR spectrum was taken in the range 4000–400 cm⁻¹ using PerkinElmer spectrometer by KBr pellet technique, while the Raman spectrum in the range 4000–10 cm⁻¹ was recorded using a laser wavelength of 514 nm with LabRAM HR (UV) spectrometer. For NLO studies, the samples were irradiated using 5ns laser pulses at the wavelength of 532 nm obtained from a frequency doubled Nd:YAG laser (Minilite I, Continuum). The sample transmission was measured by taking the samples in a 1 mm cuvette and translating along the laser beam axis (z axis) with the aid of a stepper motor-controlled translation stage.

3. Computational methods

Density functional theory (DFT) calculations were performed using the Gaussian'09 [25] program with the B3LYP functional using 6–311++G(d,p) basis set. NBO 3.1 [26] version of Gaussian '09 was employed for natural bond orbital analysis. Quantum Theory of Atoms In Molecules (AIM) was used for intramolecular interaction analysis with the assistance of AIMAll program package [27], while Hirshfeld surface analysis for intermolecular interaction study was carried out using Crystal Explorer software version 3.1 [28]. Time-dependent DFT was employed for UV–Vis studies in gas phase while the solvent effects were modelled using Polarizable Continuum Model (PCM) [29,30]. The vibrational assignments were aided by MOLVIB program [31] which gives potential energy distribution for each vibration of the system. Hyperpolarizability calculations for getting the nonlinear optical parameters were carried out using both B3LYP and range separated

CAM-B3LYP functional with the same basis set that was used for geometry optimization. Since single molecule was taken for computational studies, intermolecular interactions were neglected.

4. Results and discussion

4.1. Solubility studies

The solubility of GuHM is determined at five different temperatures 30, 35, 40, 45 and 50 °C by dissolving the solute in water and water-methanol (1:1) solvents. The saturated solution was allowed to reach equilibrium in about 6 h in an air tight container maintained at a constant temperature of 30 °C with stirring, and then the solubility was analyzed gravimetrically. The same procedure was repeated for different temperatures, and the solubility curve obtained is shown in Fig. 1(a). It is found that GuHM has better water-methanol solubility as compared to water. The positive slope of the solubility curve enables growth by either slow cooling or slow evaporation; in this work, slow evaporation method has been chosen. For growing large size crystals, the above single crystals were dissolved in water-methanol (1:1) mixture at room temperature and a supersaturated solution was obtained and then optimally closed it for controlled evaporation. Due to the evaporation of solvent, optically good quality GuHM single crystals were obtained and are washed with ethanol after a period of two weeks. The grown GuHM crystal of dimension 12 × 10 × 7 mm³ is shown in Fig. 1(b).

4.2. X-ray diffraction data collection and refinement of structure

The crystal parameters of GuHM obtained from the SCXRD experiment are presented in Table 1. Since GuHM is a new crystal, the crystallographic information file (cif) of GuHM is deposited in Cambridge Crystallographic Data Centre (CCDC) with deposition number 1863868. The crystal has orthorhombic structure with *Pbca* space group. All the non-hydrogen atoms are refined with anisotropic thermal parameters.

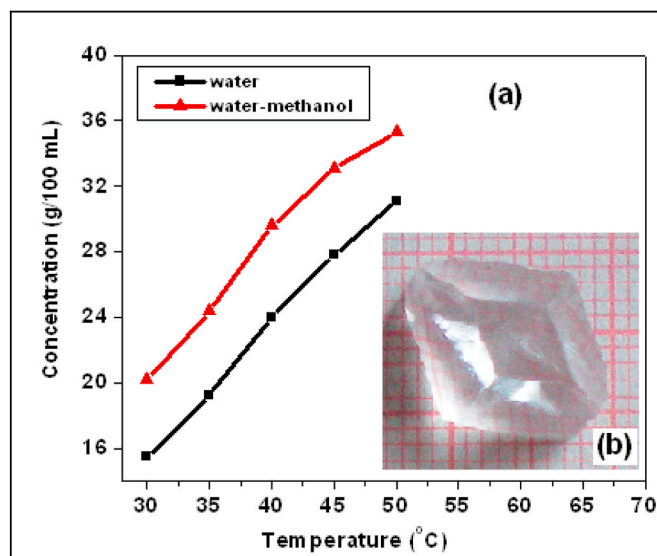


Fig. 1. (a) Solubility curve of GuHM (b) Single crystal of GuHM.

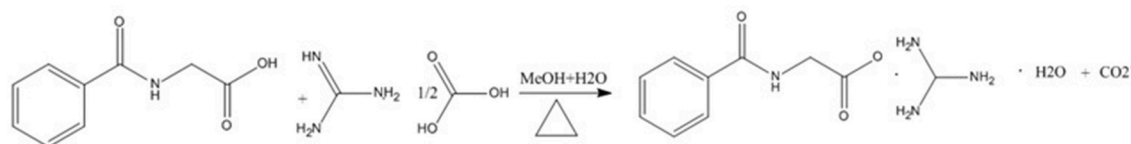


Table 1
Crystal data and structure refinement for GuHM.

Empirical formula	C ₁₀ H ₁₆ N ₄ O ₄
Formula weight	256.27
Temperature	296(2) K
Wavelength	0.71073 Å
Crystal system, Space group	Orthorhombic, P b c a
Unit cell dimensions	a = 15.3421(8) Å, b = 9.4669(5) Å, c = 17.4641(9) Å $\alpha = \beta = \gamma = 90^\circ$.
Volume	2536.5(2) Å ³
Z	8
Density (calculated)	1.342 Mg/m ³
Absorption coefficient	0.105 mm ⁻¹
F(000)	1088
Crystal size	0.500 x 0.400 x 0.200 mm ³
Theta range for data collection	3.418–24.999°
Index ranges	−18 ≤ h ≤ 18, −11 ≤ k ≤ 11, −20 ≤ l ≤ 20
Reflections collected	27,718
Independent reflections	2232 [R(int) = 0.1005]
Completeness to theta = 24.999°	99.8%
Absorption correction	Semi-empirical from equivalents
Max. And min. Transmission	0.96 and 0.93
Refinement method	Full-matrix least-squares on F ²
Data/restraints/parameters	2232/0/171
Goodness-of-fit on F ²	1.037
Final R indices [I > 2σ(I)]	R1 = 0.0438, wR2 = 0.0881
R indices (all data)	R1 = 0.0747, wR2 = 0.1007
Extinction coefficient	0.0082(11)
Largest diff. Peak and hole	0.163 and −0.143 e.Å ⁻³

The H atoms which participate in the H-bonds are identified by difference Fourier method and are refined with isotropic thermal parameters. All the other H atoms (-CH) are positioned geometrically and refined using a riding model, the corresponding data are given in the *Supplementary Information* (See [Tables S1–S4](#)). The Oak-Ridge Thermal Ellipsoid Plot (ORTEP) and the crystal packing of GuHM that are obtained from ORTEP program [32] are shown in [Fig. 2](#). During the chemical reaction of guanidine carbonate with hippuric acid, carbon dioxide is liberated and water is generated as a side product. This water molecule bridges the hippurate anions and the guanidine cations. There are no bonds observed between the water molecule and the cation or the anion except hydrogen bonds. Simply, it is an adduct of hippuric acid and guanidine compounds that are stabilized by the hydrogen bonds. In the crystal structure, the hippurate anion and the guanidinium cation link through the water molecule, which interacts with the cation and the anion through N–H...O and O–H...O hydrogen bonding respectively, to form a three-dimensional network.

The indexed powder XRD pattern that was obtained from the Mercury 3.8 program [33] using the crystallographic information file (cif) of

GuHM and the experimental powder XRD pattern of GuHM are compared in [Fig. 3](#) and both the patterns are found to match very well.

The coupled 2theta/omega scan of the GuHM crystal using the Rigaku SmartLab High Resolution X-ray diffractometer (HR-XRD) gives the Bragg peaks that are oriented in a single direction and the peak at 20.28° is subjected to omega scan which gives the rocking curve with 19.87° of FWHM as shown in [Fig. 4](#). The symmetric sharp peak with no branches and the low value of FWHM indicate the purity of the crystal with less defects.

4.3. Optimized Geometry

The structure of GuHM that was obtained from SCXRD and the one obtained using DFT are shown in [Fig. 5\(a\)–\(b\)](#) respectively. The three entities of GuHM, viz., the hippurate anion (HA), the guanidinium cation and the water molecule, co-exist as a single group by linking each other through the hydrogen bonds, that also exist between the other groups present in the crystal. The XRD structure shows that the hydrogen bonds in a single group of GuHM exists between the carboxylate oxygen O₂ of HA and the hydrogen atom (H₂₃) of the water molecule, and between the oxygen atom O₂₂ of the water molecule and the NH bonds N₃₁–H₃₂ and N₂₅–H₂₇ of the guanidinium cation. Other hydrogen bonds are also identified between the atoms of the same group and the atoms of the other groups of the crystal and they also have a key role in stabilizing the whole crystal structure. These hydrogen bonds are formed between (a) O₃ and the guanidine bonds, H₃₃–N₃₁ and H₃₀–N₂₈, (b) O₁ and the guanidine bonds, N₂₅–H₂₆ and N₂₈–H₂₉ (c) N₄–H₂₁ and O₁ and (d) O₂₂–H₂₄ and O₂.

Optimization of GuHM geometry using B3LYP and CAM-B3LYP functionals give the same structure and the molecule exists in stable state by forming hydrogen bonds within the same group. Slight deviation from the XRD structure is observed due to the absence of inter-molecular hydrogen bonding. A hydrogen bond is formed between N₃₁–H₃₂ and O₂ in the optimized structure instead of that between N₃₁–H₃₂ and O₂₂ in the XRD structure. The computational geometrical parameters of GuHM that are obtained using B3LYP functional are compared with experimental data and the tabulations are given in [Table S5 \(Supplementary Information\)](#).

4.4. Natural bond orbital analysis

The bonding interactions within the single system of optimized GuHM are depicted from the Natural Bond Orbital (NBO) analysis and the stabilization energy (E⁽²⁾) for each interaction between the filled (core orbitals, σ or π bonds, lone-pairs etc.) and unfilled (σ*, π*, Rydberg orbitals) orbitals are obtained using the second-order perturbation theory [34,35]. The natural bond orbitals, which are considered as the

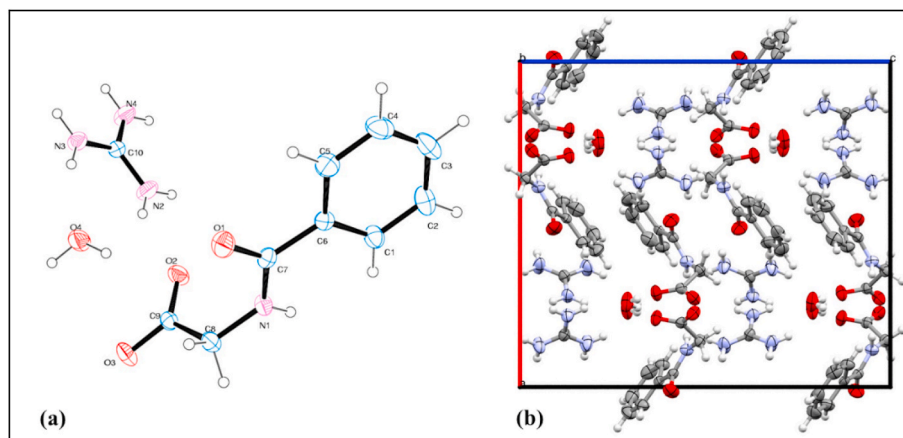


Fig. 2. (a) ORTEP of GuHM (b) Crystal packing of GuHM along b axis.

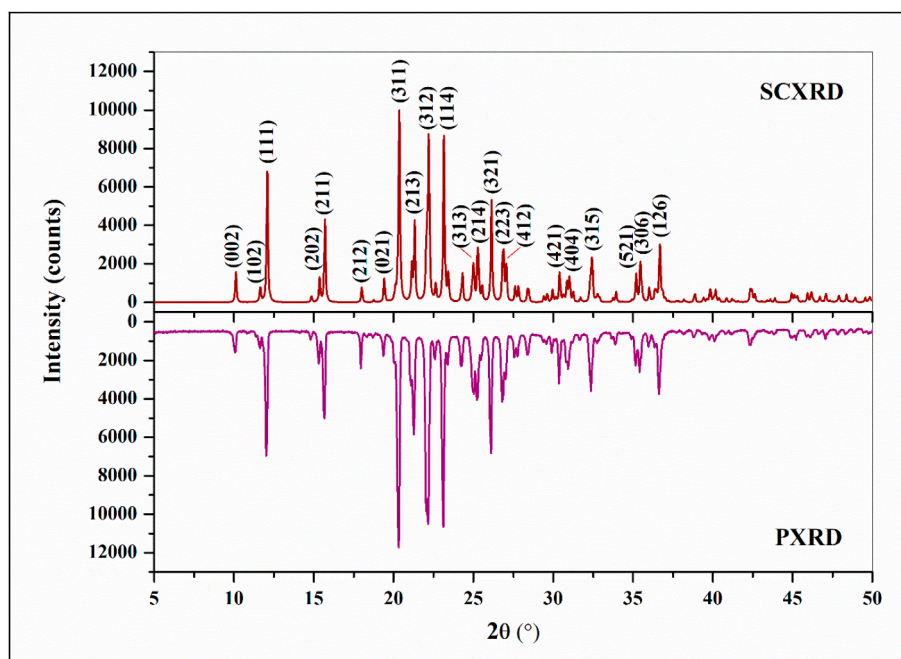


Fig. 3. Comparison of Powder XRD patterns of GuHM from Mercury software and from PXRD experiment.

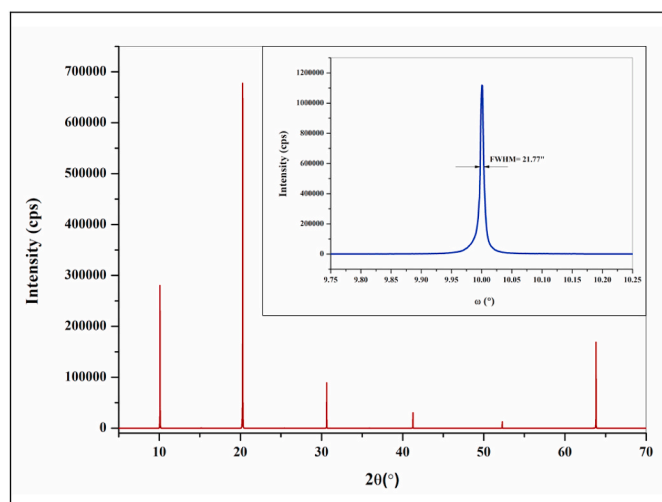


Fig. 4. HRXRD $2\theta/\omega$ scan of GuHM crystal. Inset shows the rocking curve for the peak at 20.28° .

chemist's basis set, consists of localized bonds and lone pairs as the molecular structural units. The interaction energy $E_{ij}^{(2)}$ between the donor NBO (i) and the acceptor NBO (j) due to the delocalization of

electrons has been estimated from the NBO Fock matrix and is given by:

$$E_{ij}^{(2)} = q_i \frac{\langle i | \hat{F} | j \rangle^2}{\varepsilon_j - \varepsilon_i} = q_i \frac{F(ij)^2}{\varepsilon_j - \varepsilon_i} \quad (1)$$

where q_i is the donor occupancy, $\hat{F}$ represents the Fock operator, ε_i and ε_j denote the NBO orbital energies, and $F(ij)$, the Fock matrix element.

In GuHM, as seen from Table 2, $\pi \rightarrow \pi^*$ interactions are observed in the phenyl ring and they extend to the carbonyl group whose lone pairs take part in the hyperconjugative $n \rightarrow \sigma^*$ interactions with adjacent bonds of N_3-C_7 and C_6-C_7 . The $n \rightarrow \sigma^*$ interactions are observed between the lone pairs of all the oxygen atoms of hippurate ion and the adjacent bonds, some of them being the hydrogen bonding interactions. The water molecule acts as a bridge between the hippurate anion and the guanidinium cation by forming intramolecular $O_{22}-H_{23}O_2$ and $N_{25}-H_{27}O_{22}$ hydrogen bonding with the anion and the cation respectively. Between N_4 and O_3-C_7 , an $n \rightarrow \pi^*$ interaction is observed with the largest value of stabilization energy (52.5 kcal/mol). The N-H and N-C bonds of the guanidinium ion interact with each other through the weak intramolecular $\sigma \rightarrow \sigma^*$ hyperconjugative interactions.

4.5. Quantum Theory of AIM

Using the quantum theory of AIM [36], all the information regarding an atom inside a molecule can be extracted through topological characterization of charge density, $\rho(r)$ and Laplacian of the charge density,

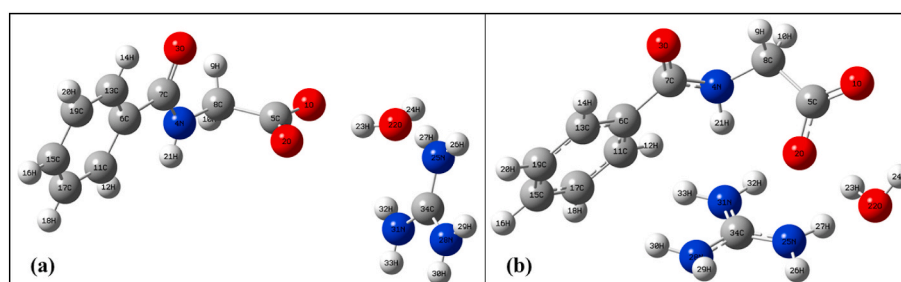


Fig. 5. (a) Structure of GuHM from SC-XRD (b) DFT optimized geometry of GuHM.

Table 2
NBO analysis of GuHM.

Donor (i)	Natural Orbital Occupancy (ED _i)	Acceptor (j)	Occupancy (ED _j)	Energy Density E(2) kcal/mol	ED _j -ED _i a.u.	F(ij) a.u
$\pi(C_6 - C_{11})$	1.64976	$\pi^*(O_3 - C_7)$	0.31713	15.74	0.31	0.063
$\pi(C_6 - C_{11})$	1.64976	$\pi^*(C_{13} - C_{19})$	0.31259	20.19	0.28	0.068
$\pi(C_6 - C_{11})$	1.64976	$\pi^*(C_{15} - C_{17})$	0.32541	19.52	0.28	0.067
$\pi(C_{13} - C_{19})$	1.65804	$\pi^*(C_6 - C_{11})$	0.36222	19.87	0.28	0.067
$\pi(C_{13} - C_{19})$	1.65804	$\pi^*(C_{15} - C_{17})$	0.32541	20.78	0.28	0.069
$\pi(C_{15} - C_{17})$	1.65238	$\pi^*(C_6 - C_{11})$	0.36222	21.02	0.28	0.069
$\pi(C_{15} - C_{17})$	1.65238	$\pi^*(C_{13} - C_{19})$	0.31259	19.27	0.28	0.066
$n_2(O_1)$	1.86548	$\sigma^*(O_2 - C_5)$	0.06999	23.38	0.73	0.119
$n_2(O_1)$	1.86548	$\sigma^*(C_5 - C_8)$	0.08945	18.94	0.60	0.096
$n_2(O_2)$	1.87984	$\sigma^*(O_1 - C_5)$	0.26078	12.03	0.85	0.091
$n_2(O_2)$	1.87984	$\sigma^*(C_5 - C_8)$	0.08945	13.94	0.64	0.085
$n_2(O_3)$	1.86779	$\sigma^*(N_4 - C_7)$	0.07036	23.42	0.72	0.118
$n_2(O_3)$	1.86779	$\sigma^*(C_6 - C_7)$	0.07077	19.29	0.65	0.102
$n_1(N_4)$	1.69347	$\pi^*(O_3 - C_7)$	0.31713	52.50	0.30	0.113
$n_2(O_2)$	1.87984	$\sigma^*(O_{22}-H_{23})$	0.05772	19.20	0.70	0.105
$n_1(O_2)$	1.93171	$\sigma^*(N_{31}-H_{32})$	0.08263	19.04	1.07	0.128
$n_2(O_{22})$	1.92760	$\sigma^*(N_{25}-H_{27})$	0.07079	30.20	0.83	0.142
$\sigma(N_{25}-H_{26})$	1.98438	$\sigma^*(N_{31}-C_{34})$	0.02130	5.47	1.18	0.072
$\sigma(N_{25}-H_{27})$	1.98357	$\sigma^*(N_{28}-C_{34})$	0.02557	5.54	1.08	0.069
$\sigma(N_{31}-H_{32})$	1.98006	$\sigma^*(N_{28}-C_{34})$	0.02557	5.92	1.07	0.071
$\sigma(N_{31}-H_{33})$	1.98380	$\sigma^*(N_{25}-C_{34})$	0.02173	5.82	1.16	0.073

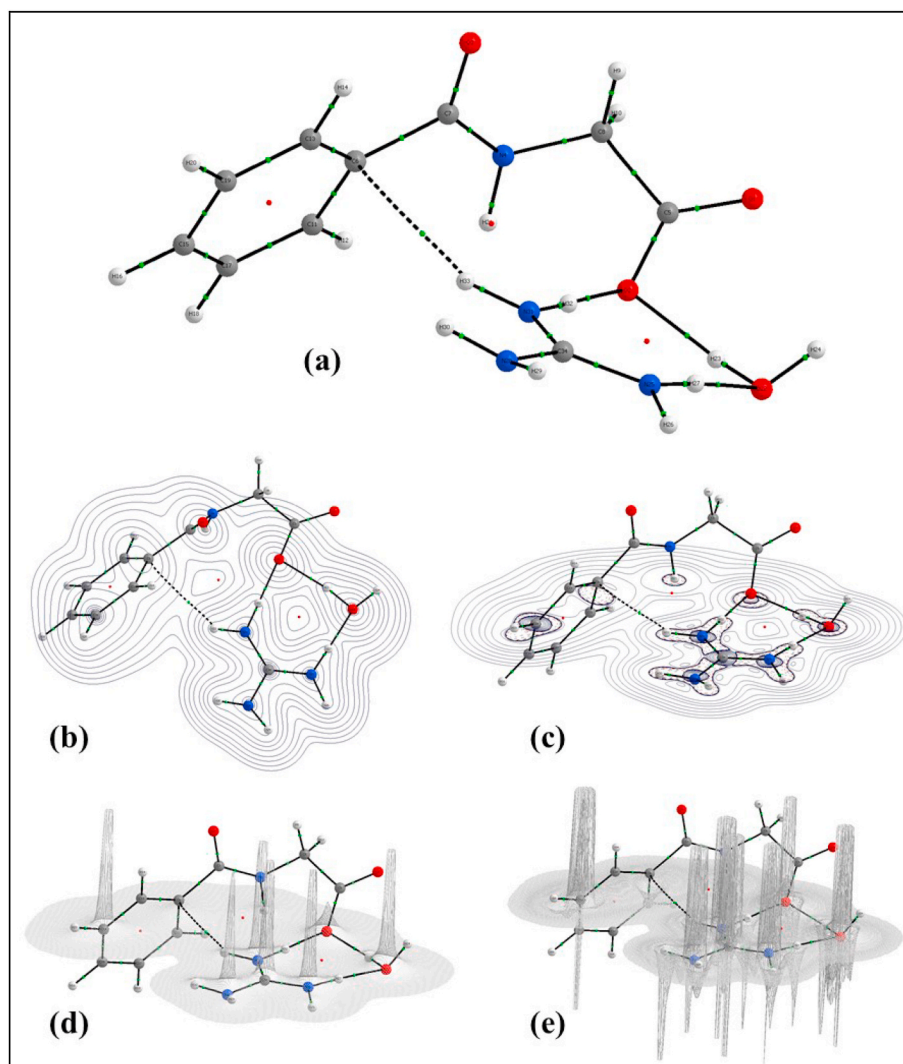


Fig. 6. (a) BCPs (green dots on the line connecting two atoms) and RCPs (red dots inside closed paths) of GuHM (b) Contour plot of charge density ρ (c) Contour plot of Laplacian of ρ (d) Relief map of ρ (e) Relief map of Laplacian of ρ . (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

$\nabla^2\rho(r)$. The critical points are those points at which the first derivative of charge density vanishes. A bond critical point (BCP) between two atoms is an indication of the formation of chemical bonds, whose nature is obtained from $\rho(r)$ and $\nabla^2\rho(r)$ values of the corresponding BCP. Similarly, the ring critical point (RCP) shows the presence of bounded ring surface and is generally found at the centre of a ring.

The values of the contours in the contour plot of charge density (Fig. 6(b)) in the plane of the three nitrogen atoms of guanidinium ion indicate the fact that the charge density has its maximum value at the nucleus and it decays as the distance from the nucleus increases. The solid and dashed contour curves in Fig. 6(c) denote the positive and the negative values of $\nabla^2\rho(r)$ respectively. A three-dimensional view of the charge density and its Laplacian is obtained from the corresponding relief maps as shown in Fig. 6(d) and (e). The relief map of ρ consists of only positive spikes while that of $\nabla^2\rho(r)$ consists of both positive and negative spikes that correspond to the positive and negative Laplacian distributions.

Several properties can be evaluated once the BCPs are determined; the charge density of the BCPs, the Hessian of the charge density, the Laplacian of the charge density and the measure of charge accumulation in terms of ellipticity, are some of them [37]. For closed shell interactions such as ionic bonds, hydrogen bonds and van der Waal's interactions, $\nabla^2\rho(r)$ should be positive and ρ should be low. Table 3 shows all the BCPs of GuHM with charge density and Laplacian of charge density of the BCPs. It can be observed that the positive values of Laplacian of ρ for GuHM exist for C₆-H₃₃ (+0.014511), O₂-H₃₂ (+0.148073), O₂-H₂₃ (+0.135292) and O₂₂-H₂₇ (+0.131646), which can be identified also from Fig. 6(a) as hydrogen bonds. From the values of ρ and Laplacian of ρ of the remaining 32 BCPs, it is found that all the other bonds are of covalent type.

Table 3
AIM analysis of GuHM.

	Bonds	ρ	$\nabla^2\rho(r)$
BCP-1	O ₂ - C ₅	0.341172	-0.497355
BCP-2	C ₅ - C ₈	0.243280	-0.552822
BCP-3	O ₁ - C ₅	0.399331	-0.300970
BCP-4	O ₃ - C ₇	0.401393	-0.318902
BCP-5	N ₄ - C ₇	0.320942	-0.936746
BCP-6	C ₈ - H ₉	0.282467	-0.970363
BCP-7	C ₆ - C ₇	0.254979	-0.612110
BCP-8	N ₄ - C ₈	0.256304	-0.648998
BCP-9	C ₈ - H ₁₀	0.277873	-0.936075
BCP-10	C ₆ - C ₁₁	0.304384	-0.831568
BCP-11	N ₄ - H ₂₁	0.341413	-1.650895
BCP-12	C ₁₁ - H ₁₂	0.282393	-0.971882
BCP-13	C ₆ - C ₁₃	0.305555	-0.842573
BCP-14	C ₁₃ - H ₁₄	0.284104	-0.987065
BCP-15	C ₆ - H ₃₃	0.005178	+0.014511
BCP-16	C ₁₁ - C ₁₇	0.308727	-0.860346
BCP-17	C ₁₅ - C ₁₉	0.308345	-0.861299
BCP-18	C ₁₃ - C ₁₉	0.309438	-0.865204
BCP-19	C ₁₅ - C ₁₇	0.309156	-0.865558
BCP-20	C ₁₅ - H ₁₆	0.281965	-0.971768
BCP-21	C ₁₇ - H ₁₈	0.282092	-0.972158
BCP-22	C ₁₉ - H ₂₀	0.281448	-0.967567
BCP-23	O ₂ - H ₃₂	0.057720	+0.148073
BCP-24	O ₂ - H ₂₃	0.047775	+0.135292
BCP-25	O ₂₂ - H ₂₃	0.323053	-2.207217
BCP-26	O ₂₂ - H ₂₄	0.360685	-2.477756
BCP-27	O ₂₂ - H ₂₇	0.049606	+0.131646
BCP-28	N ₃₁ - H ₃₂	0.288126	-1.487660
BCP-29	N ₂₅ - C ₃₄	0.347841	-1.110573
BCP-30	N ₂₅ - H ₂₆	0.339646	-1.629446
BCP-31	N ₂₅ - H ₂₇	0.296417	-1.554938
BCP-32	N ₃₁ - C ₃₄	0.353992	-1.125635
BCP-33	N ₃₁ - H ₃₃	0.337451	-1.642378
BCP-34	N ₂₈ - C ₃₄	0.323531	-1.019447
BCP-35	N ₂₈ - H ₂₉	0.339555	-1.650447
BCP-36	N ₂₈ - H ₃₀	0.339470	-1.659279

4.6. Hirshfeld Surface analysis

Hirshfeld surface is a region of space around a particular molecule whose electron density is greater than that of the other molecules in a crystal [38,39]. This surface is useful for finding the intermolecular interactions which, in turn, will help in finding the crystal packing and design of the crystal structure. A 2D graphical format of Hirshfeld surface is represented as the fingerprint plot (fpp) which envisages the types of interactions that a molecule experiences being in the bulk. Each fingerprint plot represents the fraction of points [40] on the Hirshfeld surface as a function of the pair (d_i, d_e), where d_i is the distance from the Hirshfeld surface to the nearest nuclei inside the surface and d_e is the distance from the surface to the nearest nuclei outside. Each point on the 2D graph represents a bin on the Hirshfeld surface and the colour manifests the fraction of points in each bin; blue represents a smaller number of points, green represents moderate fraction of points and red represents the highest fraction.

Fig. 7 shows a comparison of the Hirshfeld surfaces and the fingerprint plots for whole GuHM molecular group, and separately for hippurate anion (HA), guanidinium cation and the water molecule. The d_{norm} of whole GuHM shows the intermolecular interaction sites, while the interactions within the system are revealed by the separate d_{norm} surfaces of each entity.

In the d_{norm} surface of GuHM there are 11 interaction sites observed; three strong sites around O₁, one at O₂, two at O₃ for HA, one at H₂₄ for water molecule, and one each at H₂₆, H₂₉, H₃₀ and H₃₃ for guanidinium ion. These sites represent the points of interaction of one system of GuHM with the other. For the analysis of interaction sites that bind the three entities of GuHM within the same system, a separate investigation of the d_{norm} surfaces around each entity is required and the representations are given by Fig. 7(b), (c) and 7(d). The d_{norm} surface of HA reveals one more interaction site at O₂, due to the O...H hydrogen bond existing between H₂₃ of water molecule and the oxygen atoms of the carboxylate group.

For guanidinium ion, the d_{norm} surface shows two extra binding sites, one at H₂₇ and the other at H₃₂, for the NH...O interaction with the oxygen atom O₂₂ of water molecule. In the case of water molecule, in addition to the site obtained earlier at H₂₄ for the whole GuHM surface, the d_{norm} surface of water molecule shows two sites at O₂₂ for the NH...O hydrogen bonding with the two NH bonds of guanidinium, and one interaction site at H₂₃ through which the O...H hydrogen bond interacts with the HA. It can be concluded that all the NH sites of guanidine, the three atoms of the water molecule and the three oxygen atoms of HA are very reactive.

Each d_{norm} surface contributes differently to the total fingerprint plot. One of the two spikes at the lower left part of the fingerprint plot for GuHM surface, which represents the OH interactions, disappears for HA and guanidinium cation surfaces; but the only spike that is present in each case is positioned at opposite places for these ions. This is because the Hirshfeld surface acts as a donor to the hydrogen bond with water molecule in the case of HA surface, while Hirshfeld surface of guanidinium ion acts as an acceptor of electrons from the oxygen atom of water molecules. In the water molecule, both kinds of interactions exist, therefore, both spikes are simultaneously present for the surface. The fpp of water molecule is highly noticeable due to the presence of green colour, which is an indication of the strength of the bonds that exist between the molecular parts.

4.7. Thermogravimetric/differential thermal analyses

The thermogravimetric and differential thermal analyses (TG/DTA) of some guanidine salts were analyzed by Fauth [41] in 1960 while an exclusive study of guanidine derivatives was reported in 1967 [42] by Carignan and Satriana. They state that the thermal stability of a material depends on the anions and the cations that take part in the analysis. Since GuHM consists of only pure guanidinium cation, the variation in

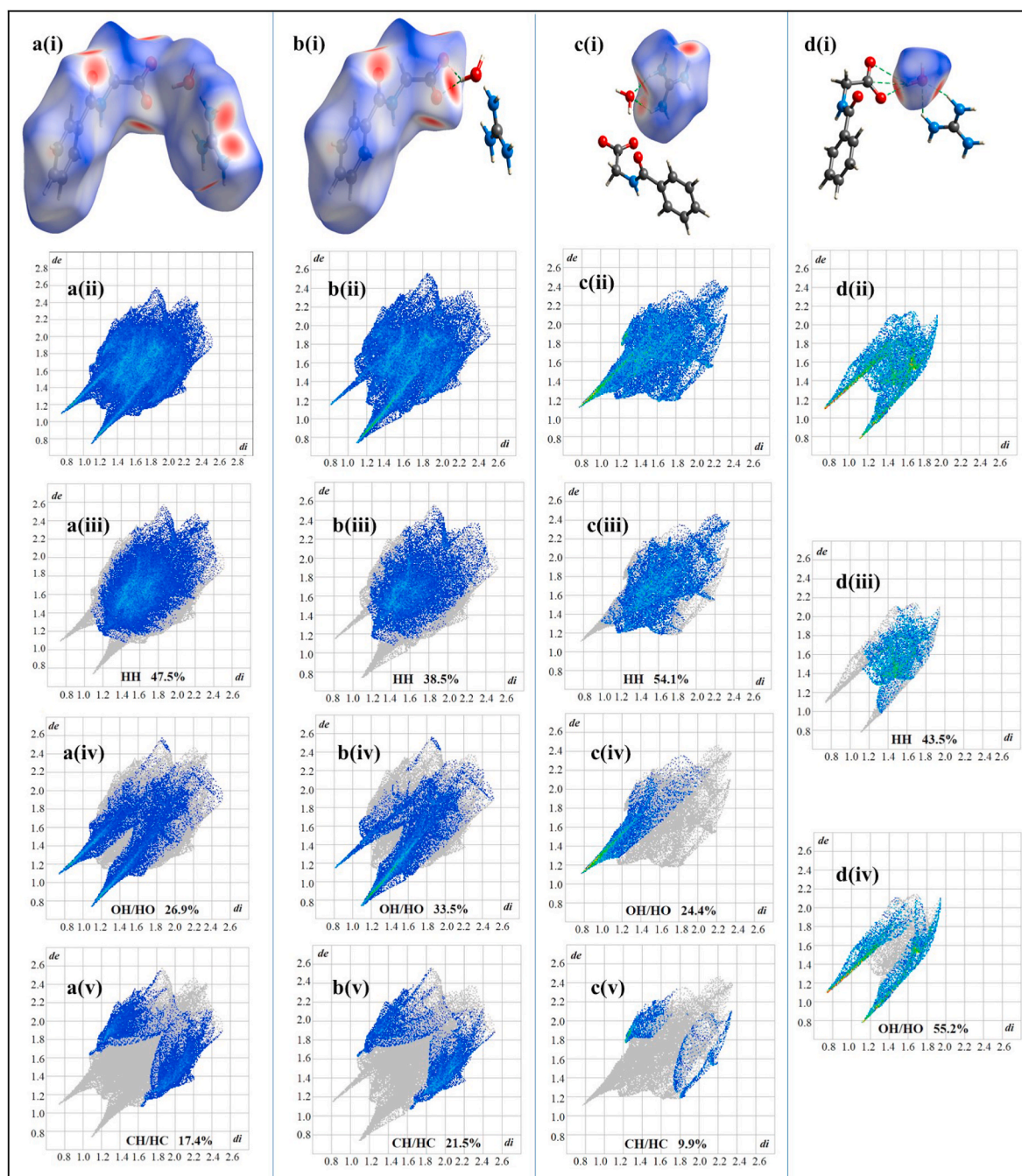


Fig. 7. Hirshfeld Surface on (a) GuHM (b) Hippurate Anion (c) Guanidinium Cation (d) Water molecule.

stability with respect to other salts is purely due to the nature of the anion, i.e. hippurate anion, and its binding to the cation. For pure hippuric acid a dip in the DTA was reported [43] at 188 °C, while slow change in mass was observed between 218 and 275 °C due to the release of carbon, oxygen and hydrogen atoms from the chain.

From the TG/DTA and the differential scanning calorimetric (DSC) curves, four levels of weight decrease (A, B, C and D in Fig. 8) are observed corresponding to the release of components of GuHM. The first decrease in the TG curve occurs around 100 °C due to the release of the water component of GuHM. The region between 161 °C and 217 °C finds the second weight loss and the corresponding endothermic peak in the DTA curve has its value at 206 °C. This stage, marked as B in the TG curve, has caused a sudden change in weight, by about almost half of the original weight. The third weight loss results in the residue weight of

14% and the peak at 318 °C may be due to the formation of peptide bond by the condensation reactions of carboxyl and amine groups [44]. The occurrence of further level of weight-decrease marked as D at 439 °C may be due to the release of all volatile gaseous components of GuHM.

4.8. Dielectric studies

The variations of dielectric permittivity and dielectric loss of GuHM crystal with frequency in the range 40 Hz to 5 MHz are shown in Fig. 9. The capacitance of GuHM crystal with flat faces is measured by varying the applied frequency and is converted into dielectric constant using the relation:

$$C = \frac{\epsilon_0 \epsilon_r A}{d} \quad (2)$$

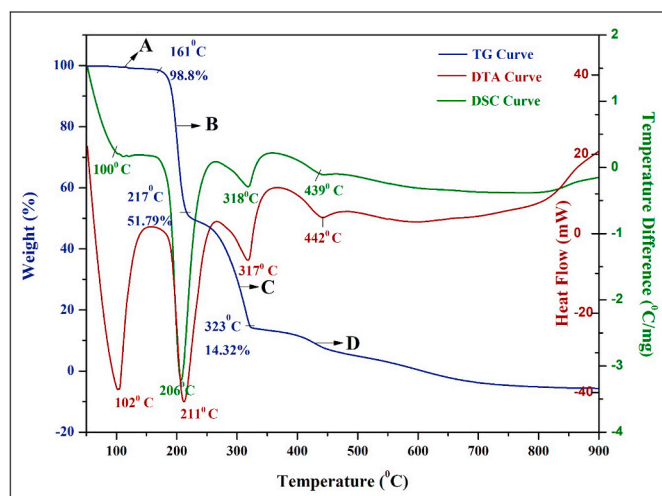


Fig. 8. TG/DTA/DSC curves of GuHM.

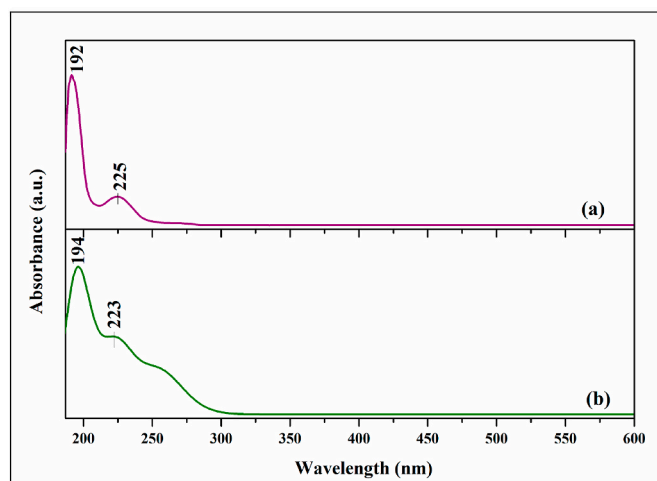


Fig. 10. UV-Visible spectra of GuHM (a) Experimental (b) Theoretical.

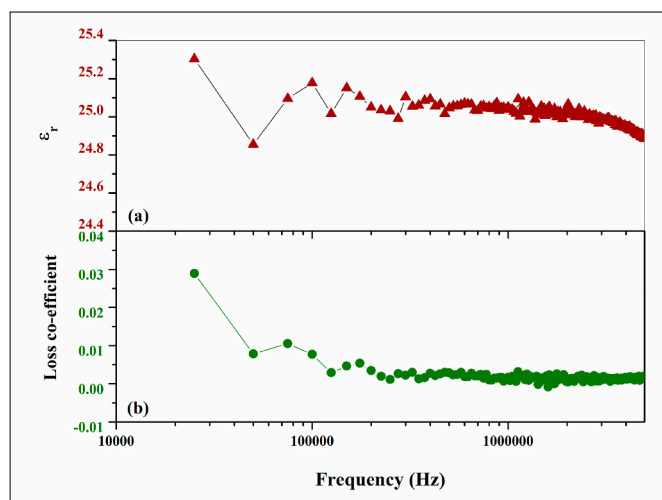


Fig. 9. (a) Dielectric permittivity vs frequency (b) Loss co-efficient vs frequency plots of GuHM.

where ϵ_0 is the permittivity of free space, ϵ_r is the relative permittivity or dielectric constant of the material, A and d are respectively the area of cross-section and the thickness of the dielectric medium (the crystal itself, in this case). It is observed that the dielectric permittivity decreases initially as the frequency increases and becomes a constant for higher frequencies, which is the normal dielectric behaviour. The variation of loss factor with frequency also behaves in a similar manner to that of dielectric constant and remains zero for larger frequencies. The electronic, ionic, dipolar, and space charge polarizations are responsible for the low value of dielectric permittivity at higher frequencies.

4.9. Linear absorption studies

Electronic absorption of molecules generally occurs in the ultraviolet (UV)-visible (vis) range of the electromagnetic spectrum. The experimental UV-vis spectrum of GuHM, in the range 187–600 nm is given in Fig. 10 and it shows that the molecule has absorption maxima at two positions in the spectrum. One peak is observed at 192 nm and the other at 225 nm, both due to the $\pi \rightarrow \pi^*$ orbital transitions of the molecule. On computing the absorption wavelengths of GuHM and tabulating (Table 4) those transitions for which the oscillator strengths are greater than 0.1 in the same wavelength range as that of the experimental

Table 4
HOMO-LUMO analysis.

Orbital Transition	Coefficient, c_i	Excitation Energy (eV)	Excitation Wavelength (nm)	Oscillator Strength, f
Excited State 1				
65 $\rightarrow$ 70	-0.16360	6.3878	194.10	0.1942
65 $\rightarrow$ 71	0.38173			
65 $\rightarrow$ 74	0.10148			
66 $\rightarrow$ 70	0.30128			
66 $\rightarrow$ 71	-0.28918			
66 $\rightarrow$ 73	-0.13245			
67 $\rightarrow$ 70	0.13018			
68 $\rightarrow$ 73	0.11463			
Excited State 2				
64 $\rightarrow$ 70	0.47261	6.3667	194.74	0.1173
65 $\rightarrow$ 69	-0.11226			
65 $\rightarrow$ 71	-0.12657			
66 $\rightarrow$ 69	0.11755			
66 $\rightarrow$ 70	0.25318			
66 $\rightarrow$ 71	0.15412			
67 $\rightarrow$ 69	0.12783			
67 $\rightarrow$ 70	-0.15892			
67 $\rightarrow$ 71	-0.15461			
Excited State 3				
64 $\rightarrow$ 69	-0.33408	5.5546	223.21	0.1083
66 $\rightarrow$ 69	-0.18688			
66 $\rightarrow$ 70	-0.14181			
67 $\rightarrow$ 69	0.24117			
68 $\rightarrow$ 70	0.50027			

spectrum, it can be seen that the theoretical values are in good agreement with the experimental values. Each absorption is a combination of different transitions whose contributions vary for different excitations. This contribution can be evaluated from the coefficient, c_i , of linear expansion of the wavefunctions of each orbital transition. The Highest Occupied Molecular Orbital (HOMO) and the Lowest Unoccupied Molecular Orbital (LUMO) are represented by the orbitals 68 and 69 respectively, and the orbital transitions in Table 4 shows that each excitation wavelength is caused by transitions not from the HOMO and the LUMO, but by other occupied and unoccupied orbitals.

4.10. Scanning Electron Microscopy/Energy Dispersive X-ray spectroscopy (SEM/EDS)

The surface morphology of powdered GuHM is shown as a Scanning Electron Microscopy (SEM) image in Fig. 11 along with the Electron Dispersive X-ray Spectrum (EDS). The weight and atomic percentages of

the atoms present in the samples are given in Table S6 (Supplementary Information).

4.11. Vibrational spectroscopic analyses

There are 96 modes of vibration for GuHM molecule, including the individual vibrations of hippurate anion, guanidinium cation and the water molecule, and also the vibrations of the bonds that connect these entities. The theoretical harmonic frequencies are scaled down uniformly [45] by a factor of 0.963 and the combined IR and Raman spectra are shown in Figs. 12 and 13. The vibrational peak assignments and the potential energy distribution (PED) using MOLVIB program are given in Table S7 (Supplementary Information).

5. Hippurate anion (HA) vibrations

Hippuric acid, with chemical formula $C_9H_9NO_3$, consists of 22 atoms with 60 normal modes of vibration. A detailed vibrational analysis of hippuric acid is reported by Karabacak et al. with a scrutiny of previous reports [8,10] also. In the case of GuHM, hippuric acid exists as an anion, forming hydrogen bond with the water molecule through the carboxylic group and with the guanidine anion through the carbonyl group. The vibrations of HA include the vibrations due to the benzene ring, the carbonyl group, the NH bond, the CH_2 group and the carboxylate group of HA.

The C–H bonds of the benzene ring and the CH_2 group contribute to the C–H stretching, in-plane bending and out-of-plane bending vibrations. The ring CH stretching vibrational spectral region is slightly above 3000 cm^{-1} , so the peaks at 3042 cm^{-1} in the FTIR and at 3080 cm^{-1} in the FT-Raman spectra are assigned to the CH stretching vibrations. The CH in-plane bending vibrations appear in the range $1000\text{--}1300\text{ cm}^{-1}$ whereas the CH out-of-plane bending vibrations are observed in the range $750\text{--}1000\text{ cm}^{-1}$. The asymmetric and symmetric stretching vibrations of CH_2 appear at 2939 and 2851 cm^{-1} in the FTIR spectrum, while only asymmetric mode is observed at 2942 cm^{-1} in the FT-Raman spectrum. The scissoring mode of CH_2 is observed as a strong peak at 1412 cm^{-1} in the FTIR and at 1382 cm^{-1} in the FT-Raman spectra.

The C=O vibration, which is a characteristic vibration of a molecule, normally causes a band in the region $1700\text{--}1800\text{ cm}^{-1}$. In GuHM, the carbonyl group of HA actively takes part in the hydrogen bonding interaction. Therefore, the corresponding wavenumber is red-shifted greatly and appears at 1654 cm^{-1} in the FTIR spectrum. The mono-substituted benzene ring has its characteristic vibrations in the range $1400\text{--}1650\text{ cm}^{-1}$ for the ring C–C stretching modes. The peaks at 1593 cm^{-1} in the IR and at 1603 cm^{-1} in the Raman spectra observed to have contributions from the CC stretching vibrations. The CN stretching mode appears at 1492 cm^{-1} in the FTIR spectrum which agrees well with the available literature [10].

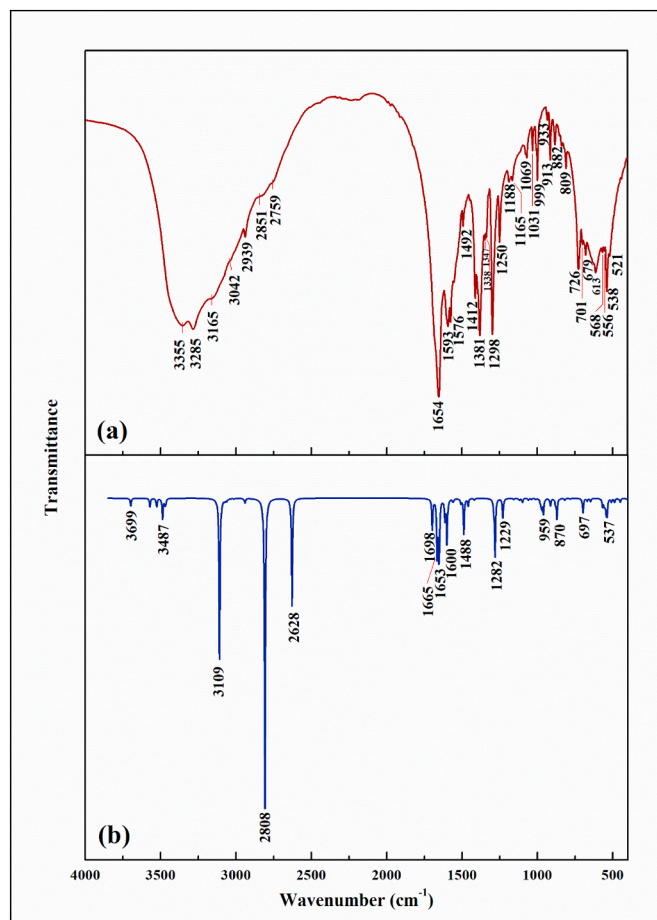


Fig. 12. Ftir spectra of GuHM (a) experimental (b) theoretical.

6. Guanidinium cation vibrations

Based on the available reports on the infra-red spectral analyses of guanidine [13] and guanidinium ion [14], the vibrational analysis of the guanidinium cation $[C(NH_2)_3]^+$ has been performed separately, which will contribute to the total vibrations of GuHM. The 24 fundamental modes of vibration of guanidinium ion are distributed among the NH_2/NH_2^+ and CN vibrational modes [46].

In the region $3500\text{--}3300\text{ cm}^{-1}$, primary amines give two bands; one corresponding to the asymmetric NH_2 stretch near 3500 cm^{-1} , and the other corresponding to the symmetric stretch near 3400 cm^{-1} [47]. The NH_2^+ asymmetric stretching vibration is expected in the region 3300 cm^{-1} while the symmetric stretching occurs in the region $2780\text{--}2600\text{ cm}^{-1}$. Since the NH_2 group of guanidinium ion participates in hydrogen

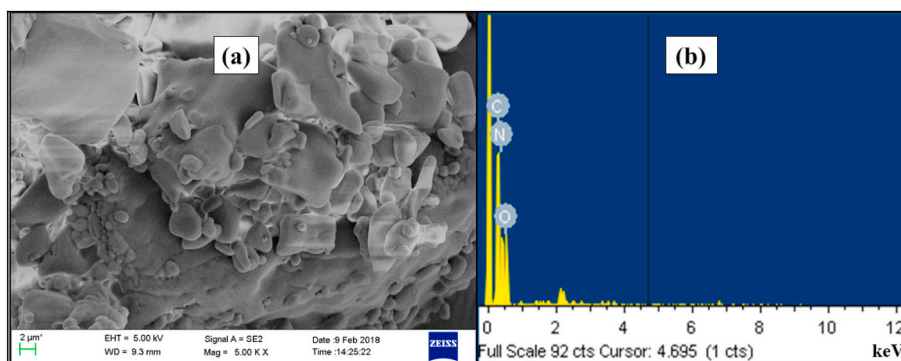


Fig. 11. SEM/EDS analysis of GuHM.

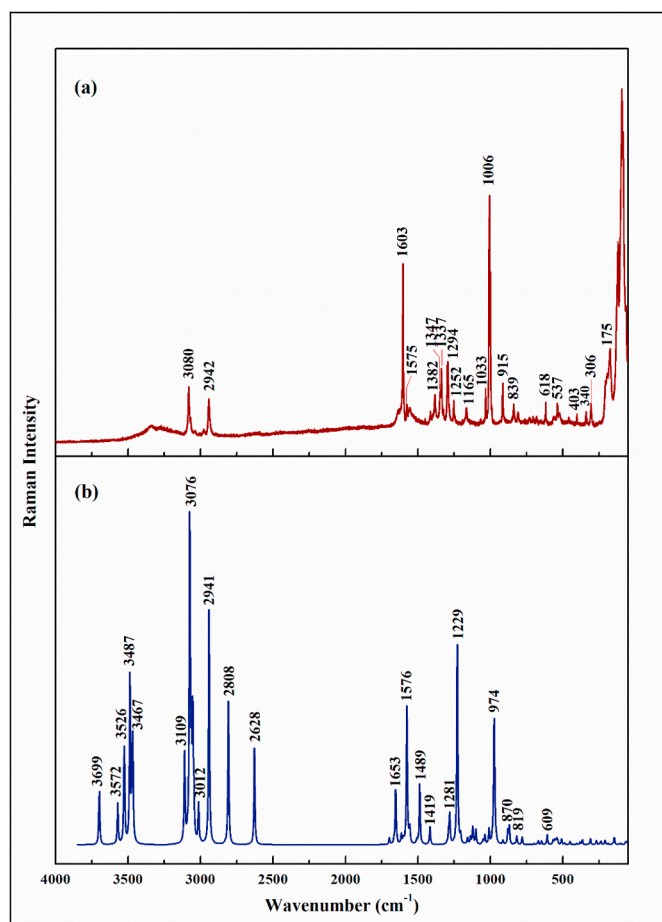


Fig. 13. FT-Raman spectra of GuHM (a) experimental (b) theoretical.

bonding, the stretching wavenumbers are lowered. The NH stretching modes of the NH_2 group of the guanidinium cation are observed as intense peaks at 3355 and 3285 cm^{-1} for the asymmetric and symmetric NH_2 stretching vibrations, while the asymmetrical stretching mode of NH_2^+ appears at 3165 cm^{-1} and the symmetrical mode is found at 2759 cm^{-1} in the FTIR spectrum. The NH_2 deformation band occurs in the region 1650–1590 cm^{-1} in the IR spectrum for primary amines [47], so the strong peak at 1654 cm^{-1} in the FTIR spectrum might have contribution from the out-of-phase $\delta(\text{NH}_2)$ deformation mode and it exhibits no Raman scattering.

Since there are two more NH_2 groups attached to the carbon atom of the $\text{C}=\text{N}$ link, it is difficult to identify the $\text{C}=\text{N}$ bond with certainty [48]. The wide range of frequency for this peak is 1718–1590 cm^{-1} and the strong peak at 1593 cm^{-1} in the FTIR and the very strong peak at 1603 cm^{-1} in the FT-Raman spectra may be attributed to the $\text{C}=\text{N}$ stretching vibration.

The peaks at 999 cm^{-1} in the FTIR and at 1006 cm^{-1} in the FT-Raman spectra are assigned to the symmetrical CN stretching mode of guanidine skeleton according to the available report [13]. The out-of-plane wagging of NH_2 is identified from the peak at 809 cm^{-1} in the FTIR spectrum, while the assignment of 726 cm^{-1} peak is done to the out-of-plane $\text{C}=\text{N}$ bending vibration. The broad feature at 645 cm^{-1} is due to the out-of-plane mode of NH groups, and the NCN rocking and deformation modes are observed at 568 and 538 cm^{-1} respectively.

7. Water molecule vibrations

For a non-hydrogen bonded water molecule, both the OH bonds are free hence the asymmetric stretching is observed above 3700 cm^{-1} , while the symmetric stretching peak lie around 3600 cm^{-1} [49]. For

instance, the vibrational modes of free water molecule have frequencies as $\nu_1 = 3652$, $\nu_2 = 1595$ and $\nu_3 = 3756 \text{ cm}^{-1}$ [50]. Due to the involvement of water molecules in the hydrogen bonding and thereby the reduction of force constant, the stretching frequencies undergo a bathochromic shift by about 300 cm^{-1} from that of the free water molecule. In the FTIR spectrum, a broadness is observed above 2750 cm^{-1} because of the presence of hydrogen bonds. The ν_2 vibration appears in the range 1400–1600 cm^{-1} , but it is difficult to assign the peak due to the presence of other molecular vibrations in this region. The theoretical scaled frequency corresponding to the ν_2 vibration is obtained at 1614 cm^{-1} .

The combined torsional oscillations of GuHM are observed throughout the spectral region, which is clear from Table S7 (Supplementary Information).

7.1. Nonlinear optical (NLO) studies

Since guanidine and hippuric acid are independently good for its nonlinear optical behaviour as per the available reports [12,16,51], it is important to check the NLO response of the combined form, GuHM. Open aperture Z-scan data of GuHM having a linear transmission of 89% is obtained by irradiating the sample with a pulsed laser of wavelength 532 nm and a laser input energy of 50 μJ . The NLO response curves of GuHM given in Fig. 14 are numerically fitted using suitable propagation equation, which suggests that the phenomenon behind the NLO behaviour is a combinational process of two-photon absorption, excited state absorption and saturable absorption. The intensity dependent effective nonlinear absorption co-efficient $\alpha(I)$ is given by the equation [52]:

$$\alpha(I) = \frac{\alpha_0}{1 + \frac{I}{I_{\text{sat}}}} + \beta I \quad (3)$$

where I and I_{sat} represent the laser input and saturation intensities respectively while α_0 and β represent the linear absorption and effective two-photon absorption co-efficients respectively. The value of β , which is a third-order NLO parameter, for GuHM is $1.98 \times 10^{-11} \text{ m/W}$ and that of I_{sat} is $50 \times 10^{11} \text{ W/m}^2$.

The computational results of polarizability and hyperpolarizability of GuHM are shown in Table 5. Except the value of anisotropic dipole polarizability, all the values corresponding to GuHM in Table 5 are larger compared to guanidine and hippuric acid individually for both B3LYP and CAM-B3LYP functionals. Thus, it is obvious that GuHM has better polarizability and hyperpolarizability which will be reflected in the nonlinear optical properties. It is clear that the $\text{N-H}\cdots\text{O}$ and $\text{O-H}\cdots\text{O}$ hydrogen bonds that exist in the compound play a major role in aiding the charge transfer between the molecular parts, thus enhancing the NLO response.

8. Conclusion

Fine quality single crystals of Guanidinium hippurate monohydrate (GuHM) were synthesized and grown using the slow evaporation solution growth technique, and its structural, thermal, and optical properties were studied. The sharpness and the small FWHM of the rocking curve of GuHM indicate very high crystallinity. The whole structure of GuHM exists in its form due to the presence of hydrogen bonds between guanidinium cation, hippurate anion, and the water molecule. In addition to the XRD analysis, the existence of hydrogen bonds is corroborated with the AIM and Hirshfeld surface analyses. The experimental and theoretical methods used for finding the NLO behaviour of GuHM confirm that the title material is a good optical limiter. Better NLO behaviour of GuHM compared to pure guanidine and hippuric acid is attributed to the presence of intra-molecular hydrogen bonding. Silver nanoparticles-added GuHM was analyzed experimentally using UV, SEM and Z-scan techniques and the results are given in Supplementary Information.

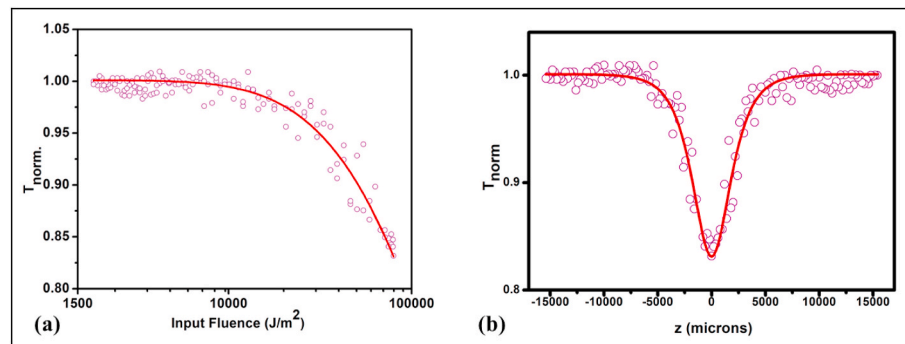


Fig. 14. Open aperture Z-scan curves of GuHM (a) Normalized transmittance vs input laser fluence (b) Normalized transmittance vs z-position. The numerical fits to the data are shown by the solid curves.

Table 5

Comparison of polarizabilities and hyperpolarizabilities of GuHM with guanidine and hippuric acid using B3LYP and CAM-B3LYP functionals.

Parameters	GuHM		Guanidine		Hippuric Acid	
	B3LYP	CAM-B3LYP	B3LYP	CAM-B3LYP	B3LYP	CAM-B3LYP
Electric Dipole Moment, μ (Debye)	11.157	11.327	2.901	2.663	4.613	4.689
Dipole polarizability, α (10^{-24} esu) $\alpha(0;0)$						
Isotropic	25.841	25.130	5.990	5.939	18.186	17.757
Anisotropic	10.651	9.861	2.732	2.661	10.903	10.337
$\alpha(-w;w)$ $w = 1059.6$ nm						
Isotropic	26.178	25.421	6.056	6.002	18.427	17.974
Anisotropic	10.957	10.096	2.781	2.704	11.191	10.585
First dipole hyperpolarizability, β (10^{-30} esu); $w = 1059.6$ nm						
$\beta(0; 0,0)$	3.769	2.017	0.195	0.247	0.243	0.071
$\beta(-w; w,0)$	4.276	2.157	0.208	0.268	0.292	0.085
$\beta(-2w; w,w)$	5.729	2.472	0.251	0.309	0.322	0.155
Second dipole hyperpolarizability, γ (10^{-36} esu); $w = 1059.6$ nm						
$\gamma(0; 0,0,0)$	10.857	6.716	1.795	1.453	5.629	4.276
$\gamma(-w; w,0,0)$	12.338	7.216	1.957	1.552	6.132	4.561
$\gamma(-2w; w,w,0)$	16.593	8.306	2.219	1.733	7.128	5.127

CRediT authorship contribution statement

Jerin Susan John: Formal analysis, Investigation, Writing - original draft. **T. Arumanayagam:** Methodology. **P. Murugakoothan:** Methodology. **D. Sajan:** Conceptualization, Writing - review & editing, Supervision. **Nithin Joy:** Resources. **Reji Philip:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.optmat.2020.110493>.

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